

ML End Sem Project Evaluation

Suicidal Sentiments Identifier

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Abstract

Social media and messaging platforms are at the centre of understanding human behaviour. Our aim is to create a machine learning model that detects suicidal tendencies in communication, typically social media posts and messages, which are in a text form. The model classifies the users through speech patterns and surrounding circumstances using a classification model, exploring methods such as Support Vector Machines, Naive Bayes, etc. The method ensures a powerful tool to save lives and intervene in the health-threatening behaviors of individuals. [GitHub](#)

1. Problem Statement Description

We understand the difficulty in understanding and identifying the suicidal tendency of a person based on the context of the text they post or use in conversation. With the increasing use of these platforms, people often express their emotional struggles and mental health issues in posts or messages. The challenge is to create a machine-learning model that can analyze this text and detect warning signs of suicidal thoughts by classifying users based on their speech patterns and contextual factors. The ultimate goal is to provide timely intervention by recognizing these tendencies and potentially saving lives. A significant challenge we face is how to adapt our machine learning model to the constantly changing language used by users, especially since the platforms we collect data from tend to be quite informal. This issue emerged as our main problem statement after the mid-semester project review.

2. Literature Survey

1. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9565132/>: This paper uses data of posts by individuals on Reddit. It uses various DL methods and ML classifiers and conducts a binary classification to detect tweets that may have suicidal tendencies.
2. https://www.researchgate.net/publication/323065338_Online_suicide_prevention_through_optimised_text_classification: This paper explores social media and investigates Major Depressive Disorder among Twitter users.

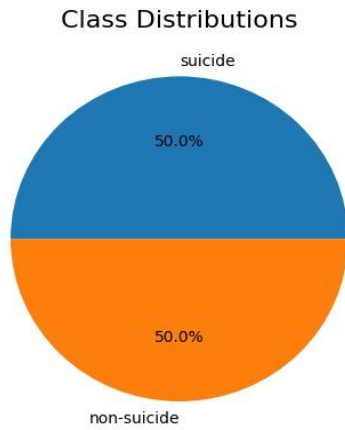
https://www.researchgate.net/publication/259948193_Predicting_Depression_via_Social_Media: This paper explores social media and investigates Major Depressive Disorder among Twitter users.

3. https://www.researchgate.net/publication/259948193_Predicting_Depression_via_Social_Media: This paper explores social media and investigates Major Depressive Disorder among Twitter users.
4. <https://floatcode.wordpress.com/2015/11/28/internet-slang-dataset/>: The dataset here consisted of multiple internet slang that are used as per November 28, 2015

3. Dataset Details

The dataset used for this project is a publicly available Reddit dataset sourced from Kaggle[2], containing approximately 230,000 posts from the **r/SuicideWatch** subreddit. These posts span a period from **December 2008 to January 2021**, covering a wide range of discussions related to suicidal tendencies and mental health support. Non-suicide posts are collected from **r/teenagers**. The dataset is evenly split between posts categorized as suicidal and non-suicidal. A key aspect of the dataset is the variation in post lengths. The average post length is **131.5 words**, with the shortest post consisting of just one word and the longest post reaching a maximum of 15,632 words. Most posts are relatively short, with a large portion (80%) containing fewer than 190 words. The length distribution is skewed, with fewer posts having longer lengths.

However, for further improvements to tackle slang words properly, we can use the [dataset](#) to prevent slang terms from being corrected into other words during spell correction. This will help in maintaining proper context.



4. Data Pre Processing

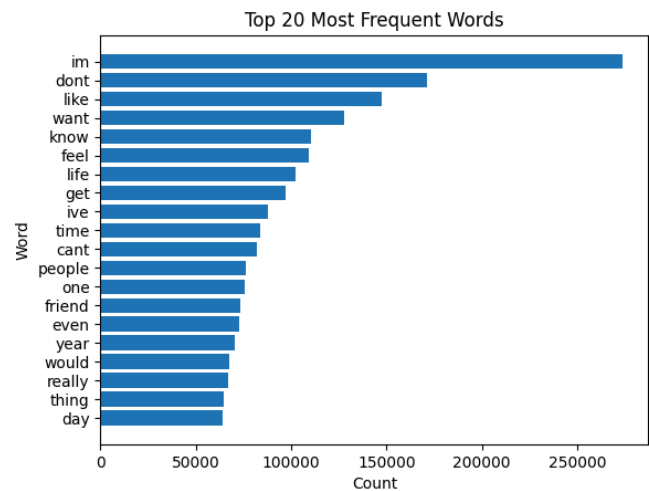
A comprehensive data preprocessing was implemented in multiple stages to clean and prepare the text data for accurate analysis and modelling. The first step involved converting all text to lowercase, ensuring uniformity by eliminating case sensitivity so that words like "Suicide" and "suicide" are treated equivalently[1]. This was followed by removing URLs, as links provide no meaningful contribution to the analysis of suicidal tendencies and can introduce unnecessary noise into the data. Numerical values, including phone numbers and underscores, were stripped from the text to avoid irrelevant information, simplifying the data for easier processing. Punctuation and special characters were also removed to clean the text further and focus solely on the meaningful content. The text was then tokenized, breaking the posts into individual words and facilitating more granular analysis. Common stopwords (such as "the," "is," and "for") that do not contribute significantly to the meaning or context of the posts were removed to emphasize more relevant words. We used Lemmatization in the hope that this would help us map the words to their root words and give us better preservation of contexts.

One problem we encountered was conjunction words. To resolve this, we used wordninja[4], a Python library, to split concatenated words into their individual component. Then, we used spellchecker to split them into proper English words. Example "helloworld" to "hello" + "world".

5. Methodology: Feature Extraction

For feature extraction, multiple techniques were selected to optimise the identification and working of the model. The primary goal of feature extraction was to simplify the data by highlighting significant terms, thereby improving model performance and interpretability. This process helps reduce noise in

the dataset and allows the model to focus on the most relevant patterns associated with suicidal thoughts. Deploying **Bag of Words (BoW)** and **Term Frequency-Inverse Document Frequency (TF-IDF)** methods for feature selection. Both techniques are widely recognised for effectively capturing important features for sentiment analysis in text data. BoW represents text based on word frequency, while TF-IDF enhances this by considering the relative importance of words across multiple posts and assigning them appropriate weights. These approaches ensure that significant terms are prioritised, improving the model's ability to detect patterns accurately. The number of features depends on the vocabulary size, which consists of all the unique words in the entire dataset (164,765 in our case).



6. Methodology: Model Selection

Since the task at hand is a binary classification problem of labelling the text as Suicidal or Non-Suicidal, we explored multiple classification algorithms to determine the best fit for this specific use case. We initially chose Logistic Regression because of its interpretability, as it provides clear coefficients indicating how each word influences the prediction of suicidal tendencies. Its straightforward nature makes it ideal for gaining insights into the model's decision-making process. We also explored Multinomial Naive Bayes because of its probabilistic nature in handling the word frequencies, as it was well suited for quick categorisation. Still, we resulted in less accuracy as it is Naive. Perceptron, on the other hand, is an iterative learning algorithm. Still, as soon as it converges, the algorithm stops and does not optimize the hyperplane further, which results in less accuracy.

Further, we used MLP with parameters `hidden_layer_sizes = (256, 16)`, `activation = 'relu'`, `solver = 'adam'`, `learning_rate_init = 0.001`, `max_iter = 100`, `batch_size = 1024`. It gave an accuracy of

92.21; however, it stopped early because the validation score was not improving over iterations. Random Forest gave an accuracy of 84.19%

7. Results and Analysis

To evaluate the models implemented, we use the models' Accuracy, Recall and F1 scores. Accuracy tells how many predictions are actually correct out of all positive/negative predictions. Recall tells us how many of all positive/negative classes we could predict. F1 score provides an outlook of classification for each case (i.e. suicidal and non-suicidal). For metrics, we use the values (TP, TN, FP, FN) obtained from the confusion matrix, which are described as follows:

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}$$

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F1 = 2 \times \frac{(Precision * Recall)}{(Precision + Recall)}$$

Accuracy: 0.8701				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.88	0.86	0.87	23287
suicide	0.86	0.88	0.87	23128
accuracy			0.87	46415
macro avg	0.87	0.87	0.87	46415
weighted avg	0.87	0.87	0.87	46415

Table 1: Performance on Perceptron Model

Accuracy: 0.8665				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.92	0.80	0.86	23287
suicide	0.82	0.93	0.87	23128
accuracy			0.87	46415
macro avg	0.87	0.87	0.87	46415
weighted avg	0.87	0.87	0.87	46415

Table 2: Performance on Multinomial Naive Bayes

Accuracy: 0.9175				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.91	0.93	0.92	23287
suicide	0.93	0.91	0.92	23128
accuracy			0.92	46415
macro avg	0.92	0.92	0.92	46415
weighted avg	0.92	0.92	0.92	46415

Table 3: Performance of Logistic Regression Model

Accuracy: 0.9221				
Classification Report:				
	precision	recall	f1-score	support
non-suicide	0.92	0.92	0.92	23287
suicide	0.92	0.92	0.92	23128
accuracy			0.92	46415
macro avg	0.92	0.92	0.92	46415
weighted avg	0.92	0.92	0.92	46415

Table 4: Performance of MLP

Model	Accuracy (%)
MultinomialNB	86.65
Perceptron	87.01
Logistic Regression	91.75
Decision Tree	84.19
Random Forest	89.36
Multi-Layer Perceptron	92.21

Each of the tables above shows the difference between the performance of each model. From Table 1, we can infer that the performance offered by the Perceptron model (accuracy = 87.01%, f1 score = 87%) lies in the middle ground of the other two models. Table 2 shows that the Multinomial Naive Bayes model has the ____ (accuracy = 86.65%, f1 score = 86% for the non-suicide class and 87% for the suicide class) and has the highest difference in non-suicide recall and suicide recall. It also has a huge difference in the precision percentage of suicide calls and non-suicide calls, which is not good as it cannot classify the non-suicide texts as accurately as the suicide texts, which can lead to more false positive alerts. Table 3 shows the performance of the Logistic Regression Model, which, through comparison, proves to be the best-performing model, achieving an accuracy of 91.75% and an F1 score of 94%. We also see the decision tree with 84.19% accuracy as the lowest random forest with 89.36% accuracy improvement from the decision tree. MLP performed the best with an accuracy of 92.21%, making it the best model.

8. Conclusion

Many people use social media as a platform to convey their feelings and troubles they are going through, or sometimes explicit suicide notes. With the increasing suicide rates, especially in India, the earlier detection of suicide intent, especially among the youth, can save lives. Our proposed model would be useful in detecting the user with such intent. Our model performs reasonably well, achieving a maximum accuracy of **92.21% by MLP**. We have implemented some of the best practices from the domain of NLP using classical Machine Learning.

During the mid-semester review, we discussed how we can explore the LLM to correct spelling mistakes and anomalies, but it turns out that when we do so, the result we get is not semantically the same as what we were expecting. We tried using free LLMs (essentially transformers, a neural network architecture that uses self-attention mechanisms to process and understand the relationships between words in a sentence, allowing for parallel processing and efficient handling of large text data), like **'google/flan-t5-small,'** ("Attention is All You Need", Vaswani et al. in 2017), but they failed to identify the sentence's meaning while trying to cure the anomaly, so they gave a poorly constructed sentence. Hence, we decided that using any LLM in the current market would not be affordable to our liking and manually iterate over such anomaly using some Python libraries. One was 'WordNinja'[4], a Python library designed to split concatenated text into individual words. Probabilistically split concatenated words using NLP based on English Wikipedia uni-gram frequencies, such as text from social media or user-generated content. This improved the dataset's quality and helped us create a more holistic and robust model for our contextual situation.

This does not prove that manual iteration is in any way a viable alternative to an automated and rather semantically robust structure; although LLM being very powerful, failed in the context, and hence the only room for improvement we can sense out is that AI Agents that are logically and contextually designed for such purposes will filter and update the linguistic changes in the most robust way possible as the social media is ever evolving. It is better to continuously update the Agent's knowledge base to update the dataset, help improve the model prediction, and have a real-time use case.

Each team member contributed equally to the project, with distinct responsibilities. Himanshu worked on the methodology and feature selection and extraction. Dhruv handled the ML model training and selection and the literature review.

Sarthak was responsible for the literature review, dataset preprocessing, and result analysis. Aayush focused on ML model selection, comparison, inference, and analysis.

References

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